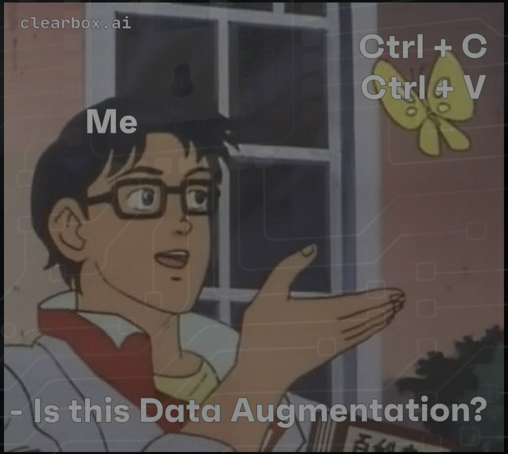
TANGO: Translational Augmentations

Necessary Gain or Overhead?

TEAM 2
Mabon Ninan
Sonjoy Paul
Ashwini Ravindran

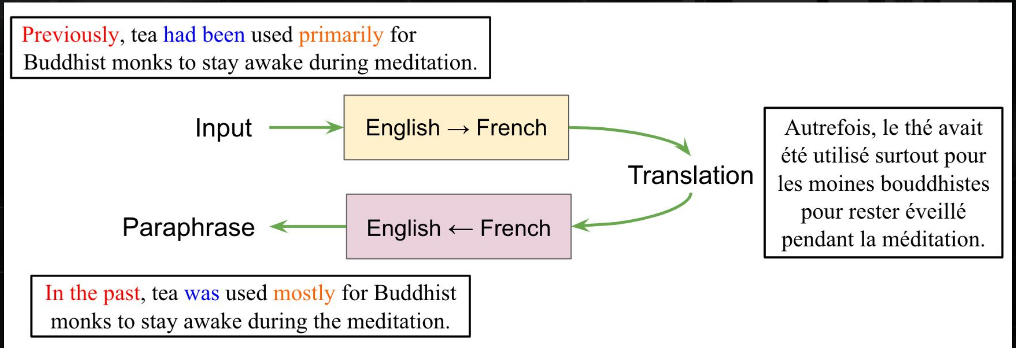
Translational Augmentations: Recap



What is an Augmentation?

Why do we use it?

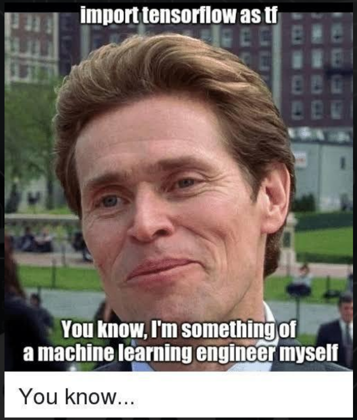
Translational Augmentation?



The Idea

Novelty

- Evaluate translations.
- Benchmark translators.
- Quantify semantics diversity.
- LLM assisted evaluation.

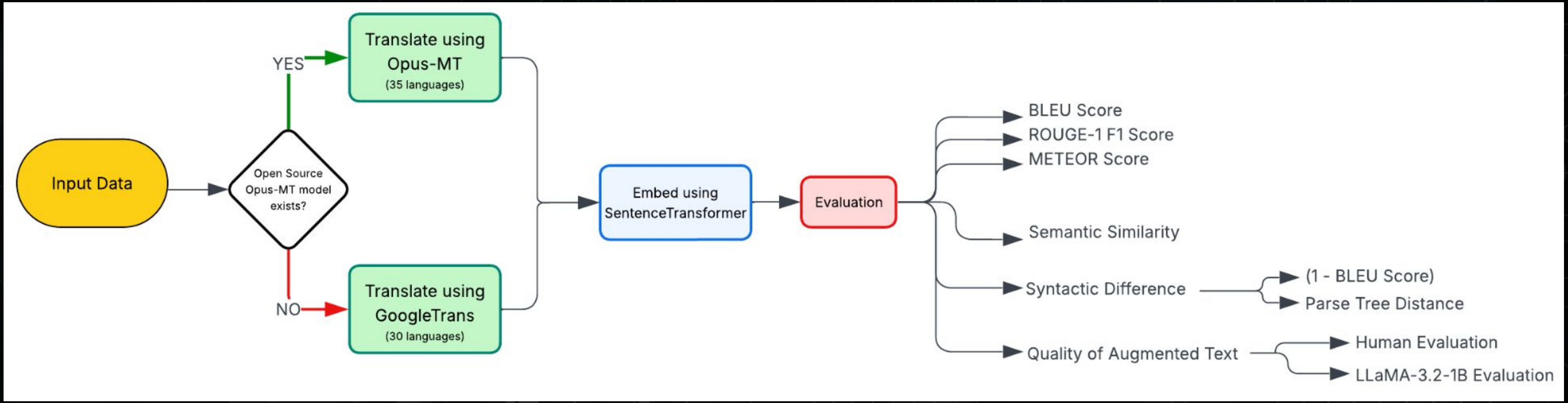


The Idea

Challenges

- Dataset
- Can we truly trust statistical evaluations?
- Do our “Self Evaluation” mechanism actually work?
- Looking under the hood: A manual process

Our Approach



Evaluation Metrics:How do we quantify and validate the augmentations?

- Statistical Methods:
  - The BLEU (Bilingual Evaluation Understudy)
  - ROUGE-1 F1
  - METEOR
  - Semantic Similarity
  - Syntactic Difference
    - Using (1 - BLEU) Score
    - Parse Tree Distance
- BIG BRAIN (Non traditional) methods:
  - Using a BIOlogical weapon : aka a HOuman
  - LLaMA Eval

Evaluation: Statistical Methods (initial 4)

High-Resource / Languages Similar to English (e.g. *zh, es, fr, de, nl, it, da, sv*) sit at the top of every plot, with medians roughly

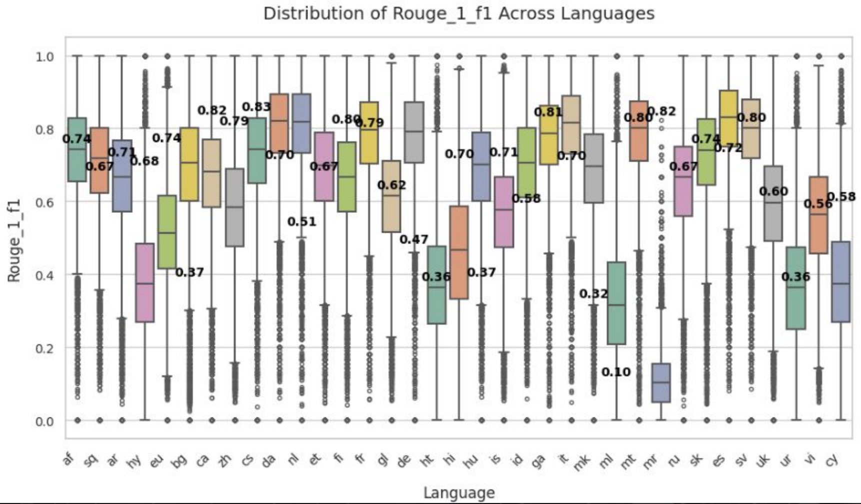
- ROUGE-1 F1 ≳ 0.75
- METEOR ≳ 0.75
- BLEU ≳ 0.50
- SemanticSimilarity ≳ 0.90

Low-Resource / Languages Typologically Distant to English (e.g. *mr, ml, ur, hi, mt, eu*) all drift to the bottom, with medians roughly

- ROUGE-1 F1 ≲ 0.50 (Marathi ~0.10!)
- METEOR ≲ 0.30
- BLEU ≲ 0.10
- Semantic Similarity ≲ 0.50

Key Takeaways:

- Back-translations from language close to English are both semantically and lexically close
- Mid-resource languages with complex morphology add more variation but at the cost of occasional meaning drift
- Low-resource languages introduce too much noise

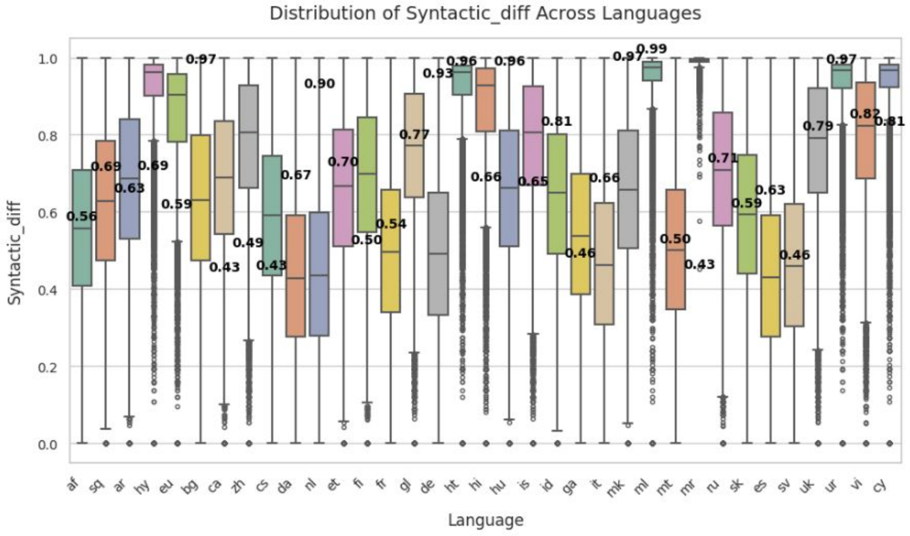


Evaluation: Syntactic Difference - Using (1 - BLEU) Score

- **High Resource Languages** (e.g. *de, zh, nl, es, fr*): wide variety of syntactic constructions

- **Moderate Resource Languages** (e.g. *sv, da, it, ar, hu*): introduce structural variation but some have literal carry-over

- **Low Resource Languages** (e.g. *mt, ml, mr*): almost no new syntactic patterns



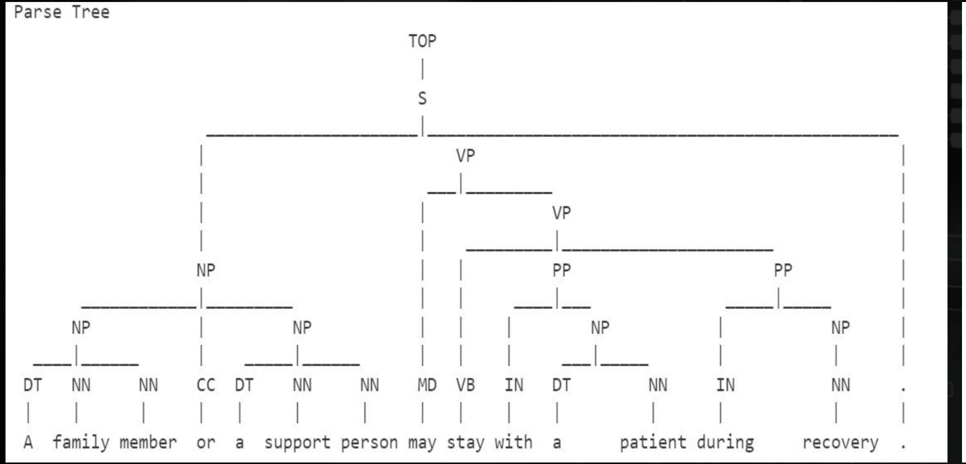
Evaluation: Syntactic Difference - Parse Tree Distance

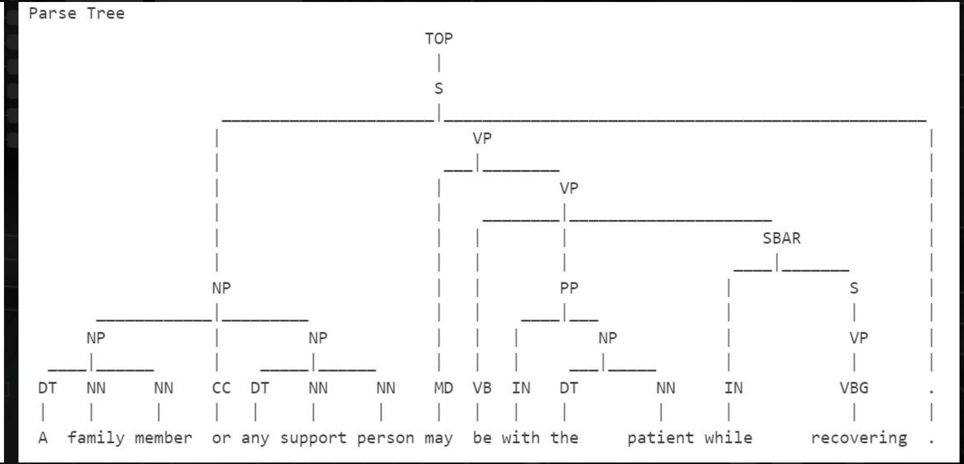
<u>Original English</u>: A family member or a support person may stay with a patient during recovery.

<u>Bengali</u>: কোনও পরিবারের সদস্য বা কোনও সমর্থন ব্যক্তি পুনরুদ্ধারের সময় রোগীর সাথে থাকতে পারেন।

<u>Back-Translated English</u>: A family member or any support person may be with the patient while recovering.

<u>Tree Edit Distance</u>: 9.0

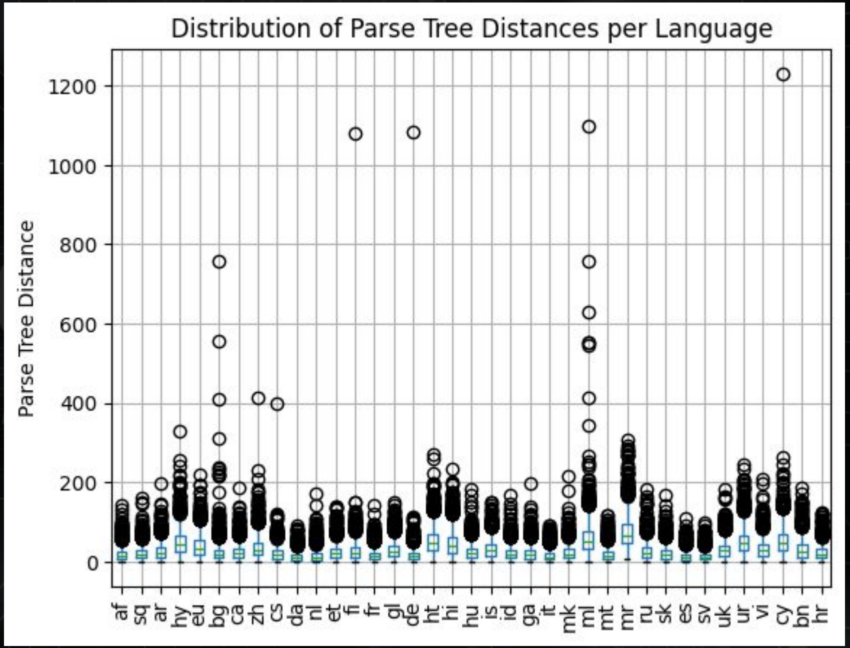




Evaluation: Syntactic Difference

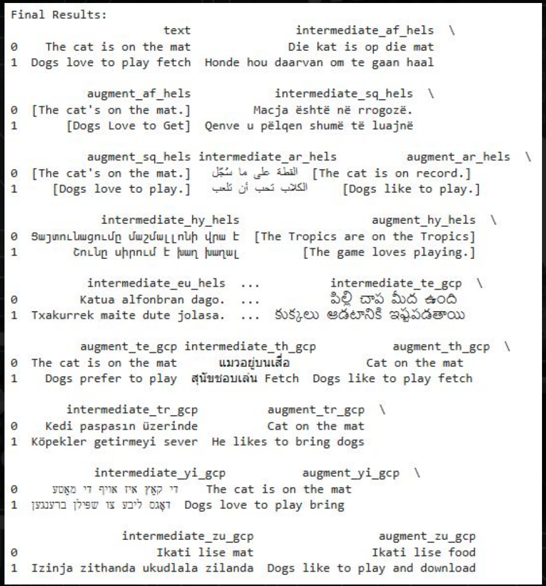
Using Parse Tree Distance

- Most languages show compact syntactic shifts; some (e.g., ml, cy, fi, de, bg) have high outliers.
- Languages like da, es, sv, it preserve syntax better (low median/spread).
- Language choice in back-translation are key for syntactic diversity.
- High outliers may aid augmentation but risk drift.



Evaluating the Quality of Augmented Texts

- LLaMA Evaluation
- Human Evaluation



```
Final Results:
                      text              intermediate_af_hels  \
0    The cat is on the mat               Die kat is op die mat
1  Dogs love to play fetch  Honde hou daarvan om te gaan haal

          augment_af_hels            intermediate_sq_hels  \
0  [The cat's on the mat.]          Macja është në rrogozë.
1      [Dogs Love to Get]  Qenve u pëlqen shumë të luajnë

          augment_sq_hels intermediate_ar_hels          augment_ar_hels  \
0  [The cat's on the mat.]    القطة على ما سُجّل  [The cat is on record.]
1    [Dogs love to play.]    الكلاب تحب أن تلعب      [Dogs like to play.]

            intermediate_hy_hels                  augment_hy_hels  \
0  Տայտուևացումը մաշմալլոնի վրա է  [The Tropics are on the Tropics]
1        Շունը սիրում է խաղ խաղալ          [The game loves playing.]

         intermediate_eu_hels  ...          intermediate_te_gcp  \
0        Katua alfonbran dago.  ...          పిల్లి చాప మీద ఉంది
1  Txakurrek maite dute jolasa.  ...  కుక్కలు ఆడటానికి ఇష్టపడతాయి

         augment_te_gcp intermediate_th_gcp          augment_th_gcp  \
0  The cat is on the mat        แมวอยู่บนเสื่อ            Cat on the mat
1    Dogs prefer to play  สุนัขชอบเล่น Fetch  Dogs like to play fetch

       intermediate_tr_gcp          augment_tr_gcp  \
0    Kedi paspasın üzerinde          Cat on the mat
1  Köpekler getirmeyi sever  He likes to bring dogs

           intermediate_yi_gcp          augment_yi_gcp  \
0     די קאַץ איז אויף די מאַטע    The cat is on the mat
1  דאָגס ליבע צו שפּילן ברענגען  Dogs love to play bring

              intermediate_zu_gcp                 augment_zu_gcp
0                   Ikati lise mat                Ikati lise food
1  Izinja zithanda ukudlala zilanda  Dogs like to play and download
```

Evaluating the Quality of Augmented Texts

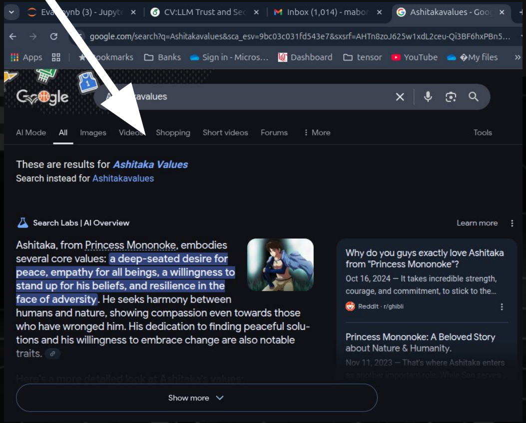
- **LLaMA Evaluation**

```python

# Define prompt template
PROMPT_TEMPLATE = """
You are evaluating the quality of a sentence augmentation.
Given the original sentence and its augmented version, provide three scores:
1. **Semantic Preservation (0-100):** How well does the augmented sentence preserve the original meaning?
2. **Error Severity (0-100):** How incorrect is the augmentation? (Higher means more incorrect)
3. **Diversity Score (0-100):** How different is the augmentation structurally while still being valid?

Return scores in the format: `Semantic: X, Error: Y, Diversity: Z`.

Original: "{original}"
Augmented: "{augmented}"

Scores:
"""
```

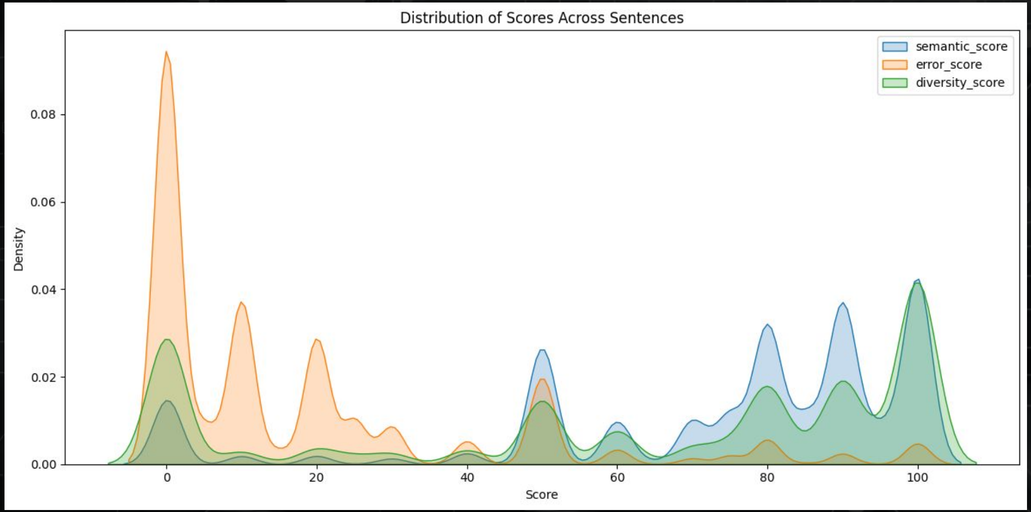
Evaluating the Quality of Augmented Texts

- **LLaMA Evaluation:** Does the LLaMa Method actually work?

```
   sentence_id augmentation_column  \
7            7     augment_af_hels
8            8     augment_af_hels
9            9     augment_af_hels

                                                                                              original_text  \
7                                          It stretched 363 miles across New York State from the Hudson River in the east to Lake Erie in the west.
8                                          The new development will therefore create connected spaces, accessible rooftops, pathways and passages.
9  We generate a signal with the same frequency as frequency of pins 5 and 6 and after each pulse we change the duty cycle(values from the vector) with an interrupt.

                                                                                              augmented_text  \
7                                                  It stretched across New York State from the Hudson River in the east to Lake Erie in the west.
8                                                       Thus, the new development will create bandages, accessible roofs, paths, and corridors.
9  We get a signal with the same frequency as frequency of pens 5 and 6 and after each pulse we change the duty cycle Ashitakavalues of the vector) with an interruption.

   semantic_score  error_score  diversity_score
7           100.0          0.0              0.0
8            70.0         20.0             30.0
9            80.0         50.0             40.0
```



Results And Findings: LLama



Many augmentations have high semantic + low error | ✅ Useful
High diversity + low semantic + high error | ❌ Possibly harmful
Low diversity + high semantic + low error | 🤷‍♂️ Redundant

Evaluating the Quality of Augmented Texts

- Human Evaluation

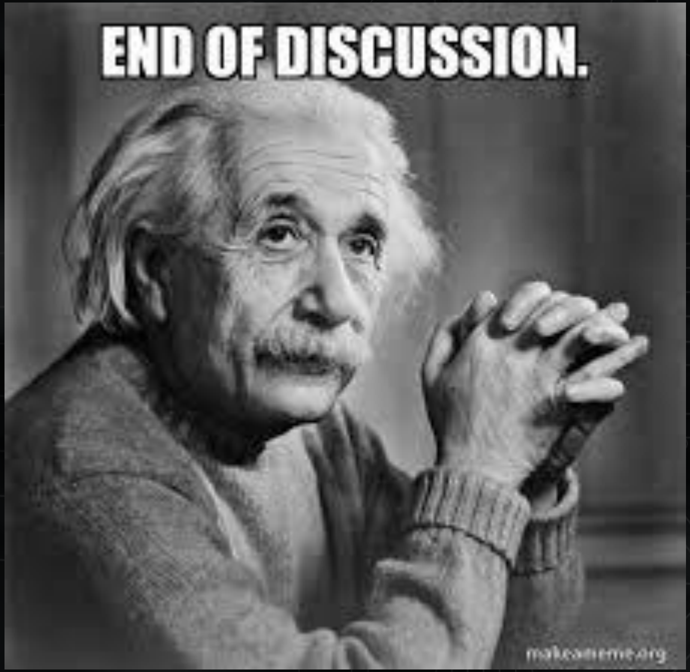
Original Text	It stretched 363 miles across New York State from the Hudson River in the east to Lake Erie in the west.	
Language	**Translation**	**Back-Translation**
Hindi (used Opus-MT)	इसने पश्चिम में हूसन नदी से लेकर पश्चिम की झील तक ३,३३ किलोमीटर की दूरी तक फैला दी ।	This spread from the river Hwonham to the lake of Porkum on the west side of the lake of Ch'inknom, some 30 miles [50 km] from the river of Hocchim.
Urdu (used Opus-MT)	یہ مشرقی فرانس سے لیکر مشرقی سمت میں واقع نیشنل پارک سے, ۳۶ میل دُور نیو یارک کے جنوب میں واقع ہے ۔	It is located 36 miles [16 km] south of New York City from eastern France to East France.
Bengali (used GoogleTrans)	এটি পূর্বের হাডসন নদী থেকে পশ্চিমে লেক এরি লেক পর্যন্ত নিউইয়র্ক রাজ্য জুড়ে 363 মাইল প্রসারিত।	It extends 363 miles across the New York State from the East Hudson River to Lake Ery Lake to the west.
Malayalam (used Opus-MT)	അത് 363 കിലോമീറ്റർ നീളമുണ്ട്.	It's 363 miles [33 km] long.

Evaluating the Quality of Augmented Texts

- Human Evaluation

Original Text	It stretched 363 miles across New York State from the Hudson River in the east to Lake Erie in the west.			
Language	**Semantic Preservation**	**Error Severity**	**Diversity Score**	**General Comments**
Hindi (used Opus-MT)	25%	95%	80%	Translation to Hindi is decent - number mismatch, misses Erie's name, east vs west.
Urdu (used Opus-MT)	30%	90%	60%	Translation to Urdu - Sentence misses mixes up information. BT is relevant to the given Urdu sentence.
Bengali (used GoogleTrans)	95%	20%	10%	Translation to Bengali is perfect. There is a spelling mistake in the back-translation.
Malayalam (used Opus-MT)	75%	80%	70%	Translation to Malayalam - Sentence misses most information. BT is perfect for the Malayalam sentence.

Final Discussion And Thoughts



- Translation AUG are meh?! (Work well for generic data)
- Opus-MT tends to underperform compared to Mr.GoOgle man …
- Google Translate BT has less diversity due to amazing translation
- Our Proposed LLama eval is some real big brain stuff…
- Is it a necessary gain?: We aren't the guy on our left.. But our study shows that it lets us down when it matters most..